Table 6. Variables Used in the Learning Gain Analysis (RQ1).

Category	Variable	Description	Scale or Range
Independent Variables	Condition	Learning condition	LLM/Web Search
	Baseline Knowledge	Pre-test knowledge score	0–15
	Vulnerability Category	Assigned vulnerability category	Injection/Memory Safety
	Age	Self-reported age group	Young/Middle/Older
	Gender	Self-identified gender	Female/Male/Non-binary
	Race	Self-identified race/ethnicity	Categorical
	Academic/Professional Status	Current academic or work role	Categorical
	Programming Level	Self-rated programming level	Beginner/Intermediate/Advanced
	Coding Experience	Years of programming experience	$<$1 / 1–2 / 2+ years
	AI Experience (General)	Frequency of general AI use	Likert (1-5)
	AI Experience (Coding)	Frequency of AI use for coding tasks	Likert (1-5)
Dependent Variables	Dimension-level Gain	Post-score $-$ Pre-score (per dimension)	Integer (-3 to $+3$)
	Overall Learning Gain	Sum of gains across all five dimensions	Integer (-15 to $+15$)
	Efficiency	Overall Learning Gain / Learning Time (z-scored)	Continuous (z-score)